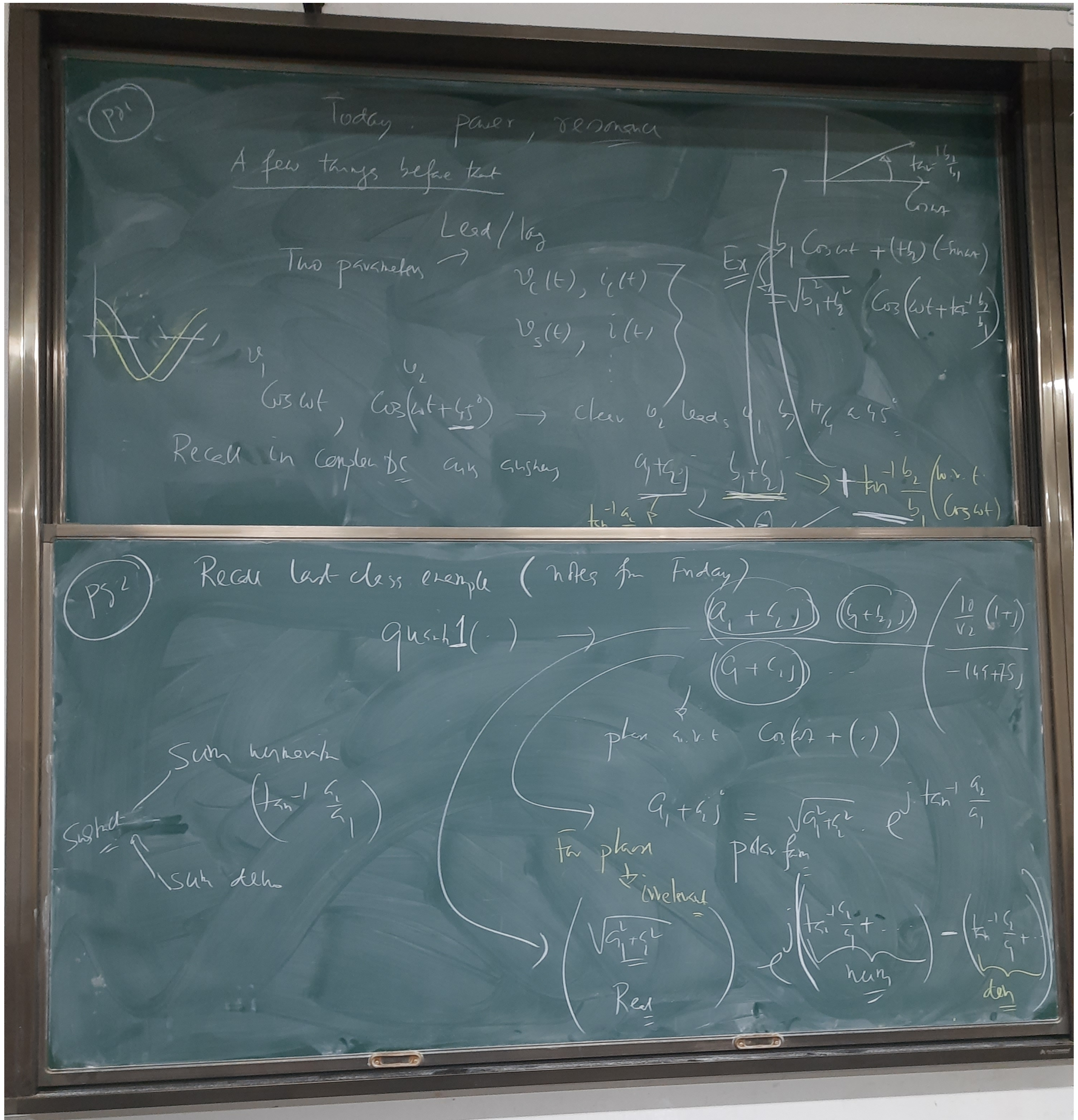


(pg 1)

Today: power, resonance

A few things before that

Two parameters → Lead / lag

$v_C(t)$, $i_C(t)$

$v_S(t)$, $i(t)$

⋮

v_1 v_2

$\cos\omega t$, $\cos(\omega t + \underline{45^\circ})$ → clear v_2 leads v_1 by $\pi/4 \approx \underline{45^\circ}$

Recall in complex DC an analogy

$\underline{a_1 + a_2 j}$, $\underline{b_1 + b_2 j}$ → $+\tan^{-1}\frac{b_2}{b_1}$ (w.r.t. $\cos\omega t$)

$\tan^{-1}\frac{a_2}{a_1}$

Ex $b_1\cos\omega t + (+b_2)(-\sin\omega t)$

$= \sqrt{b_1^2 + b_2^2}\left(\cos\left(\omega t + \tan^{-1}\frac{b_2}{b_1}\right)\right)$

$\tan^{-1}\frac{b_2}{b_1}$

$\cos\omega t$

(pg 2)

Recall last class example (notes from Friday)

question 1(.) → $\dfrac{\boxed{a_1 + a_2 j}\ \boxed{b_1 + b_2 j}}{\boxed{c_1 + c_2 j}\ \cdots}$ $\left(\dfrac{\frac{10}{\sqrt{2}}(1+j)}{-(45+75j)}\right)$

phase w.r.t. $\cos(\omega t + (\cdot))$

Sum numerator

$\left(\tan^{-1}\frac{a_2}{a_1}\right)$

Subtract

Sum den

$a_1 + a_2 j = \sqrt{a_1^2 + a_2^2}\, e^{j\tan^{-1}\frac{a_2}{a_1}}$

for phase polar form

↓ irrelevant

$\left(\underline{\sqrt{a_1^2 + a_2^2}}\ \ \text{Real}\right)$ $e^{j\left(\underbrace{\tan^{-1}\frac{a_2}{a_1} + \cdots}_{\text{num}}\right) - \left(\underbrace{\tan^{-1}\frac{c_2}{c_1} + \cdots}_{\text{den}}\right)}$

PS3

quant(2) $\dashrightarrow$ $\dfrac{(a_1 + a_1)\,(\;\ldots\;)}{(d_1 + d_1)\ldots}$

$\dfrac{\text{qnt 1}}{\text{qnt2}} = \dfrac{>(\;\;)\;(\;\;)}{\cancel{\;}(\;\;)\;(\;\;}$

Calculate w.r.t. cos ωt
phasor

+ve ⇒ qn 1 leads q. [illegible]

Notations in phasors

$\sim \rightarrow \dfrac{V_1 \cos \omega t + V_2 (-\sin \omega t)}{\hookrightarrow\; V_1 + jV_2}$

$\sqrt{V_1^2 + V_2^2} \swarrow V\angle\phi \searrow \tan^{-1}\dfrac{V_2}{V_1}$

Impedance / Reactance

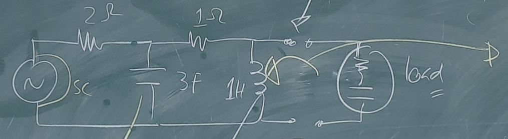
V_{oc} in AC / complex

Complex resistance

$\downarrow$

Thevenin Impedance

$\omega = 5$



$\downarrow$

$\dfrac{V}{\sqrt{2}} + j\dfrac{V}{\sqrt{2}}$

$\downarrow$

$\dfrac{V}{\sqrt{2}} \cos \overset{5}{\omega} t - \dfrac{V}{\sqrt{2}} \sin \overset{5}{\omega} t$

$-\dfrac{j}{5.3}$

$j \cdot 1.5$

Thevenin / Norton

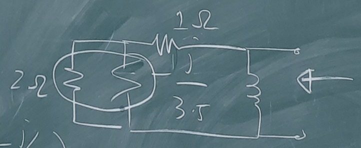
$Z_{th} = R_{th} + j\,X_{th}$

resistance

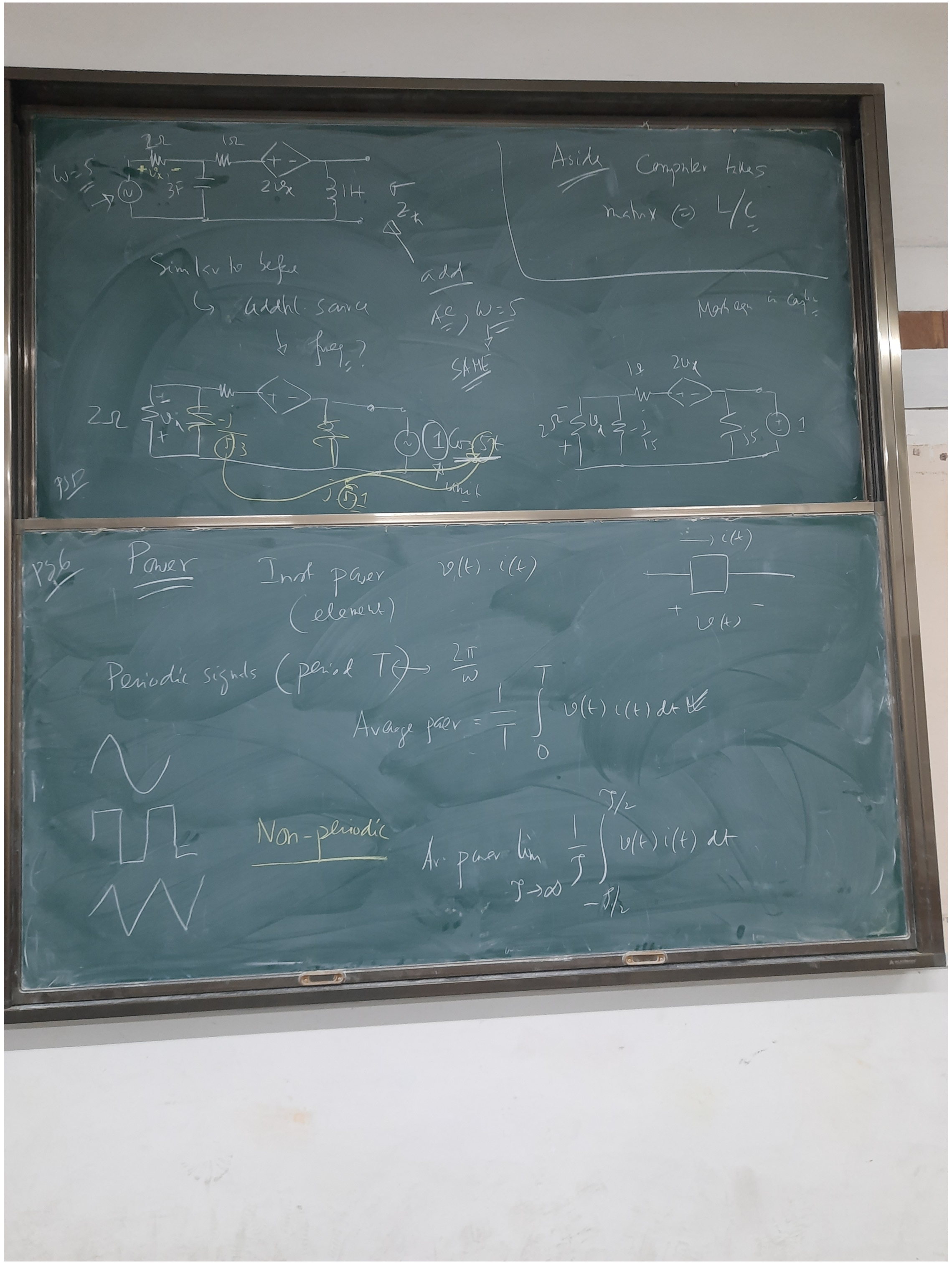
(R)

reactance

(L + C)



parallel fn $\left(\dfrac{2\,(-j/15)}{2 - j/15} + 1\Omega\right) \parallel j5$



Similar to before

↳ addtl. source

↳ freq?

add

AC, $\omega = 5$

SAME

Aside Computer takes matrix ⟺ L/C

Matrix eqn in Caylu

PS5

PS6

Power

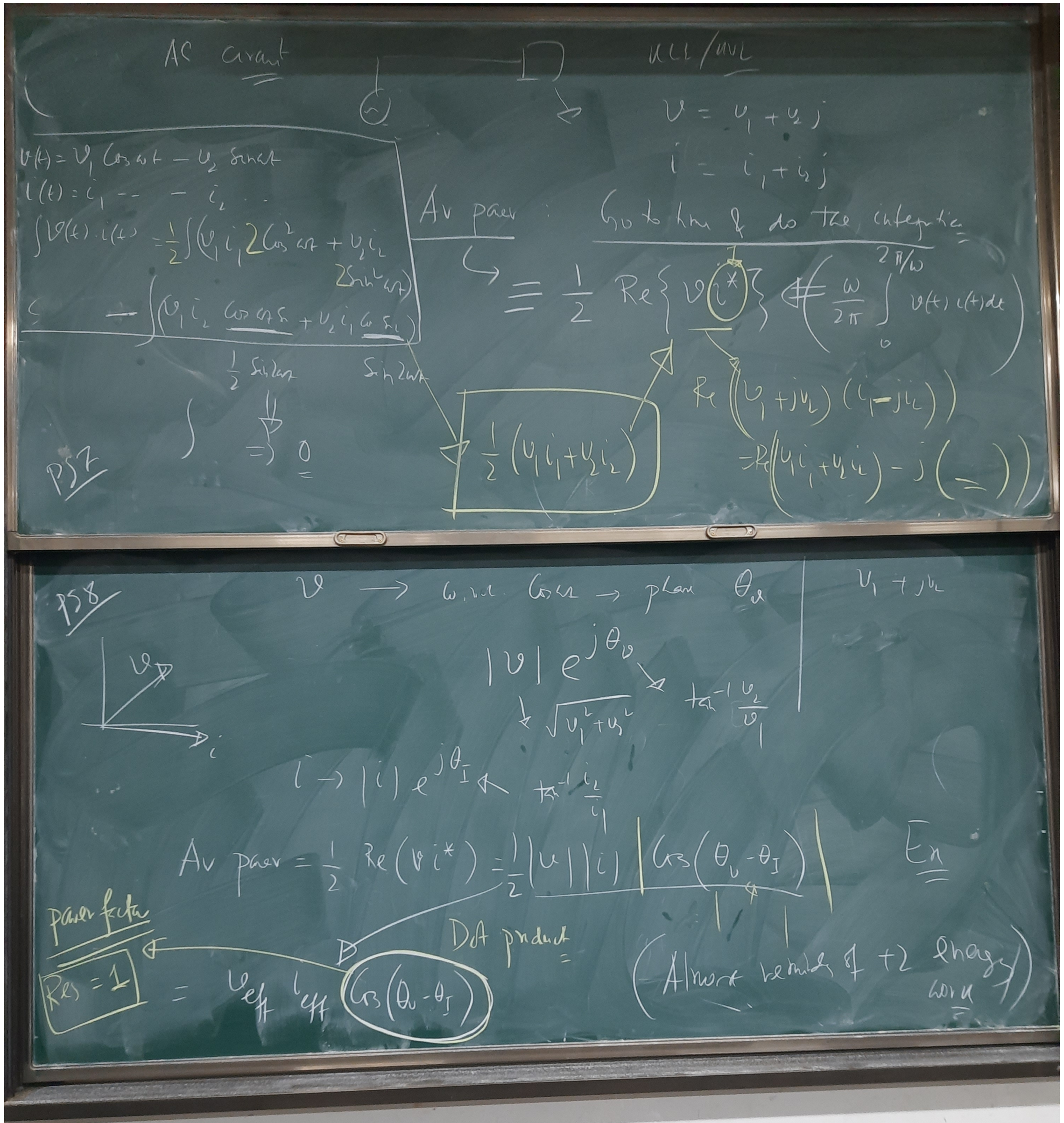
Inst power (element) $v(t) \cdot i(t)$

Periodic signals (period T) $\rightarrow \frac{2\pi}{\omega}$

$$\text{Average power} = \frac{1}{T} \int_0^T v(t)\, i(t)\, dt$$

Non-periodic

$$\text{Av. power} \lim_{\mathcal{T} \to \infty} \frac{1}{\mathcal{T}} \int_{-\mathcal{T}/2}^{\mathcal{T}/2} v(t)\, i(t)\, dt$$



AC circuit

KCL / KVL

$v = v_1 + v_2 j$

$i = i_1 + i_2 j$

$v(t) = v_1 \cos\omega t - v_2 \sin\omega t$

$i(t) = i_1 \cdots - i_2 \cdots$

$\int v(t) \cdot i(t) = \frac{1}{2}\int \left(v_1 i_1 2\cos^2\omega t + v_2 i_2 2\sin^2\omega t \right)$

$- \int \left(v_1 i_2 \underline{\cos\omega t \sin} + v_2 i_1 \underline{\cos \sin} \right)$

$\frac{1}{2}\sin 2\omega t$ $\quad \sin 2\omega t$

$\int \Rightarrow \underline{\underline{0}}$

PS7

Av power : Go to time & do the integration =

$$\equiv \frac{1}{2} \mathrm{Re}\left\{ v\, i^{*} \right\} \quad \left(\frac{\omega}{2\pi} \int_0^{2\pi/\omega} v(t)\, i(t)\, dt \right)$$

$$\boxed{\frac{1}{2}\left(v_1 i_1 + v_2 i_2 \right)}$$

$\mathrm{Re}\left((v_1 + j v_2)(i_1 - j i_2) \right)$

$= \mathrm{Re}\left((v_1 i_1 + v_2 i_2) - j(\underline{\quad}) \right)$

PS8

$v \longrightarrow$ w.r.t. $\cos\omega t \rightarrow$ phase θ_v

$v_1 + j v_2$

$|v|\, e^{j\theta_v}$

$\sqrt{v_1^2 + v_2^2}$ $\quad \tan^{-1}\frac{v_2}{v_1}$

$i \rightarrow |i|\, e^{j\theta_I}$ $\quad \tan^{-1}\frac{i_2}{i_1}$

$$\text{Av power} = \frac{1}{2}\mathrm{Re}(v i^{*}) = \frac{1}{2}|v||i| \left| \cos(\theta_v - \theta_I) \right|$$

Ex

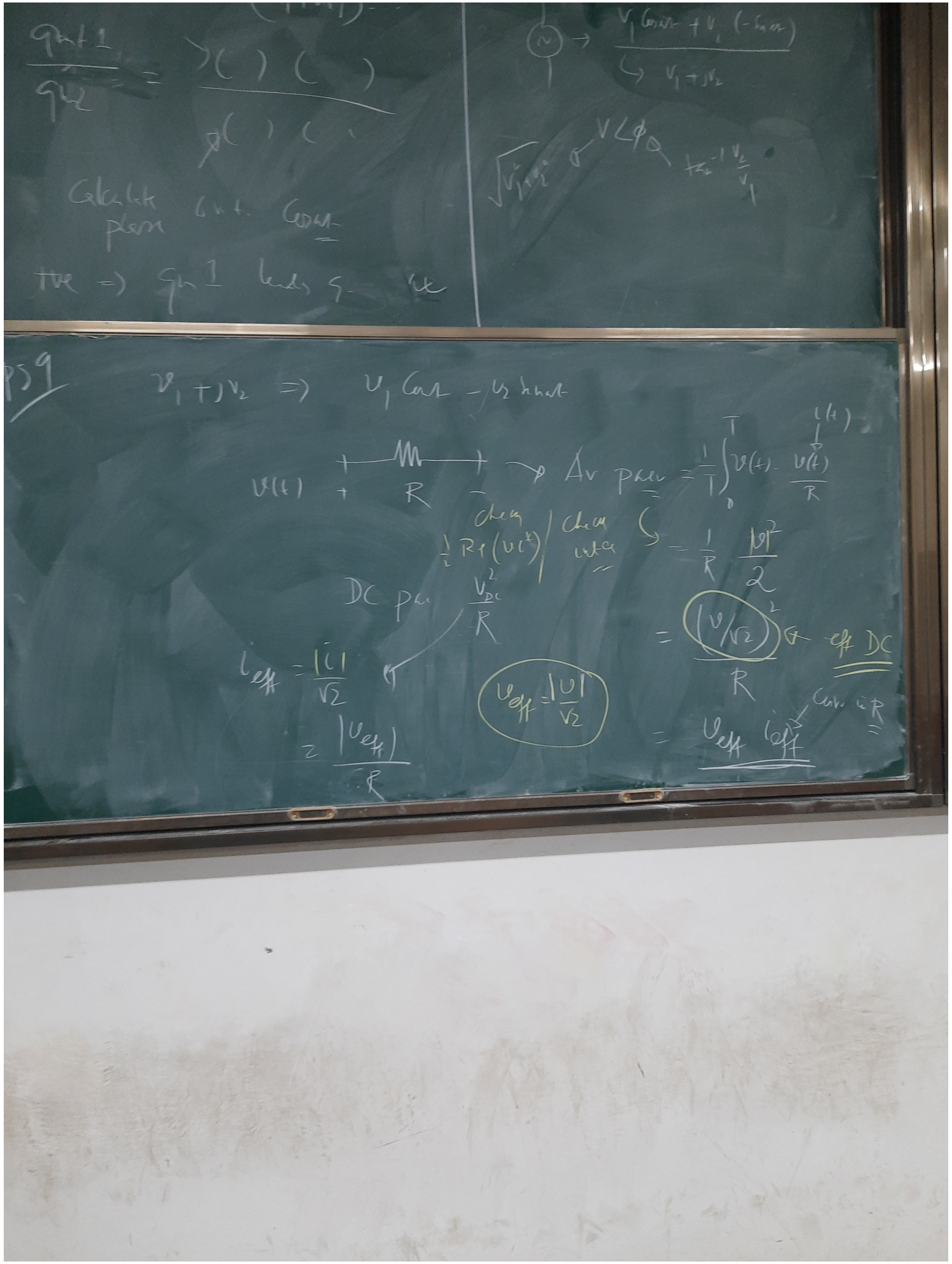
Dot product

(Almost reminds of +2 charge work)

power factor

$\boxed{\mathrm{Res} = 1}$

$= v_{eff}\, i_{eff} \cos(\theta_v - \theta_I)$



$$\frac{q_{n}1}{q_{n}2} = \frac{\gamma(\ )(\ )}{\cancel{\gamma}(\ )(\ )}$$

Calculate phasor const. Const

+ve ⇒ qn 1 leads q... [illegible]

$\rightarrow \overline{V_1\cos\omega t + V_2(-\sin\omega t)}$

↪ $V_1 + jV_2$

$\sqrt{V_1^2+V_2^2}$ $V\angle\phi$ $\tan^{-1}\frac{V_2}{V_1}$

PS9

$V_1 + jV_2 \Rightarrow V_1\cos\omega t - V_2\sin\omega t$

$v(t)$ + R − → Av power $= \frac{1}{T}\int_0^T v(t)\cdot\frac{\overset{i(t)}{v(t)}}{R}$

check $\frac{1}{2}\mathrm{Re}(vi^*)$ | check integ

↪ $= \frac{1}{R}\frac{|v|^2}{2}$

DC pow. $\frac{V_{DC}^2}{R}$

$= \frac{\boxed{(v/\sqrt{2})}^2}{R}$ ← eff DC

$i_{eff} = \frac{|i|}{\sqrt{2}} = \frac{|v_{eff}|}{R}$

$\boxed{v_{eff} = \frac{|v|}{\sqrt{2}}}$

$= v_{eff}\, i_{eff}$

curr. in R